

HUGO, a High-Resolution Tactile Emulator for Complex Surfaces

Yair Herbst

Mechanical Engineering, Technion -
Israel Institute of Technology
yair.herbst@technion.ac.il

Alon Wolf

Mechanical Engineering, Technion -
Israel Institute of Technology
alonw@technion.ac.il

Lihi Zelnik-Manor

Electrical and Computer Engineering,
Technion - Israel Institute of
Technology
lihi@technion.ac.il

ABSTRACT

Many of our activities rely on tactile feedback perceived through mechanoreceptors in our skin. While visual and auditory devices provide immersive experiences, cutaneous feedback devices are typically limited in the range of sensations they provide and are hence usually used and tested on relatively simple synthetic surfaces. In this paper we demonstrate HUGO, a device designed in a human-centered process, triggering the mechanoreceptors sensitive to pressure, low-frequency vibrations, and high-frequency vibrations, enabling one to experience touch of surfaces “in-the-wild”. The device is based on a parallel manipulator and a pin-array, that operate simultaneously at 200Hz and emulate coarse and fine geometrical features, respectively. The decomposition into coarse and fine features, alongside the high operation frequency, enable simulation of virtual surfaces. HUGO will be showcased in multiple applications such as social interactions, e-commerce and gaming to allow haptic interaction with real-world surfaces.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction devices; Haptic devices; Human computer interaction (HCI); Human computer interaction (HCI); HCI design and evaluation methods; User studies.

KEYWORDS

Haptics, High-Resolution Haptics, Haptic Textures, Human Computer Interface, User Study

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1 INTRODUCTION

While touch is involved in almost all activities that we do in the real world, in the virtual world touch is not yet sufficiently available. Being able to touch objects virtually could be useful in numerous domains. For example, in e-commerce touch would enable customers to feel the products they are buying, which could provide a better

and more enjoyable shopping experience as well as reduce returns of unsatisfactory items. Such a device will need to convey the sensation of fine textures and by doing so, feel and choose different materials. A virtual touch device can also be easily integrated in the vision of the metaverse to generate a more immersive experience involving multiple senses. Such a device should enable users to interact with objects as if they were real. To do so it will need to provide multiple types of sensations, including shear, normal forces, macro and micro roughness and warmth.

Human tactile perception is based on four types of mechanoreceptors lying in different depths relative to the skin surface. The combination of these four types of mechanoreceptors creates the known dimensionalities of mechanical perception: hardness, friction, macro roughness and fine roughness [3, 7]. Previously proposed tactile feedback devices were typically aimed at solving one aspect of the human haptic perception. For example, a device that generates normal forces at different orientations to emulate surface hardness [1], normal and shear forces that simulate friction [6], or a tactile pin-array that simulates fine roughness vibrations [9]. While these devices perform well in their intended setting, they are not capable of creating immersive experiences since they do not integrate multiple modalities. Recent studies have discussed and suggested multi-modal devices capable of conveying two or more aspects of the tactile perception [8, 10]. While these multi-modal devices offer a significant leap towards an immersive haptic interface, many focused only on material properties without attention to geometric structure.

The main contribution of this paper is HUGO, a High-resolution UX for Geometrical Objects (Figure 1). HUGO can emulate surfaces in-the-wild, i.e., the surfaces can have complex geometry such as surfaces used for computer games or reconstructed from real images of objects or products. The key idea behind HUGO’s design is to stimulate multiple types of mechanoreceptors, and through that achieve the ability to emulate multiple tactile modalities, including hardness, macro roughness and fine roughness.

Reviewing previous studies, we observe that most of them perform empirical evaluations on relatively simple synthetic surfaces, such as surfaces showing lines in different orientations and thicknesses. This is not surprising given the single modality nature of these devices. Others perform more extensive evaluations of material identification, and some worked on higher resolution synthetic surfaces such as digits and letters. We wish to push forward the baseline of tactile device evaluation, towards realistic surfaces “in-the-wild”.

HUGO’s performance was evaluated in an experiment (IRB approved) where 20 subjects were challenged in two scenarios, (1) a set of synthesized surfaces divided into pairs to evaluate resolution and feedback capabilities using controllable parameters. Users were

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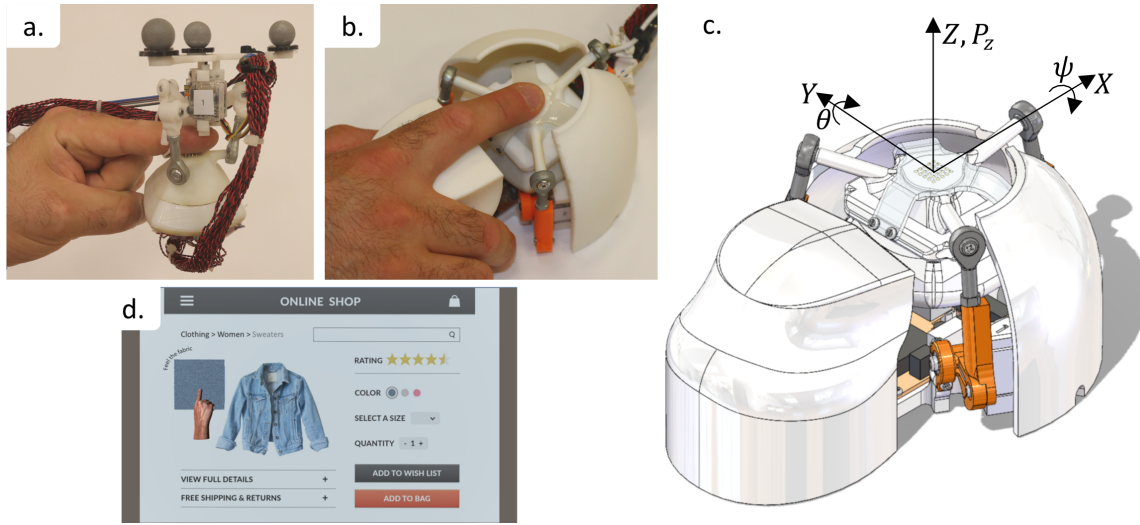


Figure 1: The HUGO haptic feedback device. a. Wearable version, b. mouse-based device, and c. its CAD model with the relevant coordinate system. d. The user interface with the hand avatar, in this example the interface shows a demo of an e-commerce application.

asked to state if the two surfaces in each pair are identical or not; (2) a set of 21 real-world surfaces divided into seven categories which are taken from an online PBR library to test surface recognition on real-world surfaces. Users were asked to match the three virtual surfaces in each category to images of those surfaces. In addition the users were asked to fill out the USE questionnaire [5] to evaluate usability, satisfaction and ease of use. The average success rate for the synthesized surfaces is 83.41% and for the real-world surfaces is 86.25%. These experiments demonstrate and quantify the capabilities of the device to emulate high-resolution (temporal and spatial) real-world surfaces.

In summary, this paper presents HUGO, a device capable of mechanically emulating multiple mechanoreceptors and provide immersive tactile feedback of surfaces in-the-wild. HUGO can be used in a plethora of applications ranging from e-commerce and gaming to sensory rehabilitation. Moreover, due to its high-resolution, the device can be used as a research tool for research on tactile sensation and feedback.

2 THE HUGO DEVICE = HIGH-RESOLUTION UX FOR GEOMETRIC OBJECTS

2.1 Hardware Implementation

HUGO was designed with the goal of enabling sensing of any geometric object, or in other words, allowing simulation of real-world surfaces. The design is based on three main key ideas that facilitate this ambitious goal. The first key idea is to rely on reverse engineering of the human somatosensory system. Specifically, the device needs to generate multiple tactile modalities out of those that can be sensed by our cutaneous mechanoreceptors. The presented design of HUGO can generate forces and vibrations at multiple frequencies, up to 200 Hz. The second key idea is to decompose the surface geometry into two scales, coarse and fine, in line with the human ability to sense both macro and micro roughness. Therefore,

HUGO's design is composed of two main components; (1) a macro geometry emulator based on a Three Revolute-Revolute-Spherical (3RRS) parallel manipulator similar to the device offered by Chinello et al. [1] with three Degrees of Freedom (DoF) – two rotations and one translation. Using these DoF the device can emulate the macro surface geometry of the object, and (2) the micro geometry and fine texture emulator which is based on a 25 DoF high speed solenoid array with pins. The 25 DoF manipulator is mounted as the platform of the 3RRS manipulator such that the pin array changes height and orientation. Lastly, to allow simulation of surfaces at very high resolution, HUGO is mounted on a computer mouse, thus allowing continuous scanning of the surface geometry by moving the finger along the object shown on the screen. The scanning motion, joint with high-frequency activation of the device, enables HUGO to overcome the spatial resolution limitation of its mechanical design, and provide a very high-resolution experience to the user. The idea is inspired by the super-resolution task in images, where multiple images are taken consequently with small misalignments and are then combined to create a single higher resolution image. Similarly, HUGO allows the user to sample the digital surface at small increments as they move their finger along the surface and by doing so collect additional information and construct an object representation in memory with a higher resolution. This concept is similar to the “window” concept in finger exploration for object recognition presented in [4]. The high accuracy tracking capabilities of the mouse and the high temporal frequency of communication (motor commands are updated at 110Hz) are critical factors that enable this super-resolution effect. The actuators are laid out in a 5X5 grid with a 2mm pitch resulting in an 10X10mm tactile area. The chosen pitch is based on our design constraints and previous studies on human ability to discriminate between two distinct points, that show mixed results but range around 1-2mm [2]. The entire pin array was covered by a 0.6mm thick silicone rubber (Shore 20A) to

increase comfort for the user and to smooth out the sensation of the distinct pins.

2.2 Simulating Geometric Objects with HUGO

Touching a virtual geometric object with HUGO requires actively tuning the 3RRS manipulator and the pin-array such that they represent the surface at the point of touch. This boils down to setting the height and orientation of the 3RRS manipulator, and the height (solenoid position) and vibrations of each of the 25 pins. The interface was designed in Unity 3D game engine, the user sees an avatar of their finger as shown in Figure 1, and controls the avatar's location by moving their finger in a scanning motion along the object surface. An overview of the mapping from a 3D surface to a sensation given to the user is shown in Figure 2.

The height of the manipulator is determined by the surface height at the specific location of the avatar on the object surface, i.e., it is assumed that the finger is laid directly on the surface. The only exception is with small crevices and holes in which the finger cannot fit and, in those cases, the finger rests on the highest areas of the surface it fits in. The orientation is determined by ray-casting a 5X5 grid, with similar dimensions to the pin array, to obtain the 25 corresponding surface normal vectors. The vectors are smoothed and averaged to generate a single normal vector that represents the macro geometry of the surface and determines the commands to the 3RRS manipulator. The averaging implies that the manipulator simulates geometry with a spatial resolution of approximately 10X10mm. To determine the height setting of each pin we ray-cast a 5X5 grid again but with the local finger orientation set according to the computed orientation of the manipulator. The nearest collision of each ray with the surface determines the height of the corresponding pin. The height of the pins generates the sensation of the local micro geometry. To simulate higher surface resolution, and provide a sensation of texture, we further consider the finger motion. As the finger moves along the surface, we update the pins height to vibrate at frequency up to 200 Hz, that is determined by the speed of motion. In addition, each pin vibrates with some small amplitude, around the position determined according to the surface geometry. The amplitude is proportional to the surface roughness which is ideally determined by a known roughness map of the specific surface being emulated.

3 SYSTEM EVALUATION

User studies were conducted to evaluate the performance of the device, the setup is shown in Figure 3. Subjects were asked to sit at a table. Their right-hand was placed on the device such that their index finger was situated on the tactile area of the device. The device was placed in a box to obscure any visual information and subjects wore protective earmuffs to obscure auditory information from the motors and pins. The subject had two screens visible, the first screen displayed the user interface of the device and the second displayed instructions for the current step of the experiment. The experiment was conducted in multiple steps; (1) calibration, user tried the system and tuned the gains of the feedback in the range of [0.5, 1.5]. (2) training, users were given sets of virtual surfaces they could feel thorough the device, some of them were visible and some were not, the users were asked to identify the

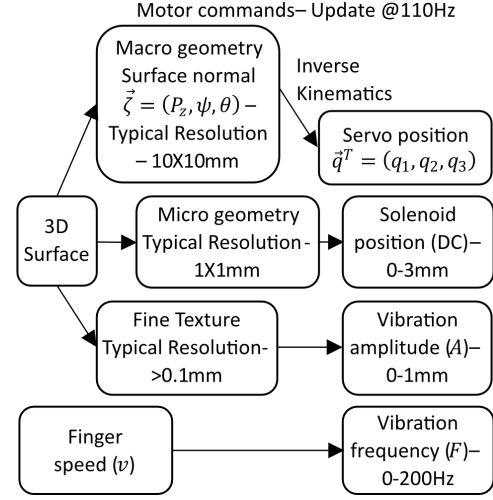


Figure 2: Mapping of 3D surfaces to motor commands and sensations by the user.



Figure 3: Experimental setup (calibration step), subject holds the device with their dominant hand. The device is placed in a box to obscure any visual information. subject wears protective earmuffs to obscure auditory information. First screen (right screen) shows system interface, second screen (left screen) shows instructions.

hidden surfaces. (3) Same Not same, users were given pairs of virtual synthesized surfaces they could feel with the device, one was visible and one hidden, the users were asked to determine whether the surfaces were identical or not. (4) Real-world surfaces, users were given 10 groups of three surfaces created from PBR materials taken from textures.com. These surfaces are based on real-world 3D scans containing high-resolution information. The surfaces were hidden but users could feel them with device and were asked to match each surface to its corresponding rendering. (5) Usability, users were asked to fill out the USE questionnaire [5] to evaluate usability, satisfaction and ease of use. The average success rate for the synthesized surfaces is 83.41% and for the real-world surfaces is 86.25%. These experiments demonstrate and quantify the capabilities of the device to emulate high-resolution real-world surfaces.

4 USER EXPERIENCE

The device will be demonstrated using a similar setup to the one described above but without obscuring visual and auditory information. Users will hold the device to feel virtual surfaces displayed either using virtual reality goggles or a regular computer monitor according to their preferences. The device will be used to explore the different objects and to feel their geometric structure. Users can choose from different interactions ranging from the real-world surface recognition task performed in the experiments to more applicable interactions such as feeling different materials for shopping online and helping Goldilocks choose the most pleasant bed for her.

5 CONCLUSIONS

In this paper we present HUGO, a device capable of conveying the sensations of complex real-world surfaces. The device is based on a simple and scalable technology and was designed according to the mechanoreceptors in the human skin. The device was validated by a set of experiments, (1) synthetic surfaces to gain understanding about the technical capabilities of the device, (2) differentiation between real-world surfaces and (3) a usability questionnaire. In all the experimental steps, users were able to achieve satisfactory results, with recognition accuracy comparable to previous studies that were conducted on sets of simple synthetic surfaces or synthesized 2D outlines.

We believe that HUGO presents a significant leap forward by, to the best of our knowledge, allowing sensing of digital complex real-world surfaces for the first time. Users were excited about working with the device and highly enjoyed the interaction. Future generations of the device can be used in a plethora of applications such as e-commerce by conveying the sensation of varied materials and structures of assorted products, it can be used as a controller with high-resolution haptics for gaming and as a tool for sensory rehabilitation. The device has been shown to experts in those fields

who were highly enthusiastic about its capabilities, and research collaborations have been initiated. Moreover, due to the wide range of sensations that could be given to the user, the device can be used as a research tool for studying tactile sensing and perception.

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